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ontrol patterns for enterprise agentic AI

siness Case — POCs 01–06

monstrations on synthetic data. Every figure in this deck is ILLUSTRATIVE — drawn from runs recorded in the public repository, quoted wit

GAI-Engineering/poc-learning-lab · MIT (code) · CC BY 4.0 (content) · October 2026

B → $24.5B

rise agentic AI market, 2024 → 2030 (Grand View Research) — 46.2% CAGR. The spend is arriving whether the controls are ready or not.

of enterprise apps

ture task-specific AI agents by the end of 2026 (Gartner, August 2025) — up from less than 5% in 2025.

cancelled

ntic AI projects will be cancelled by the end of 2027 (Gartner, June 2025) — cost, unclear value, inadequate risk controls.

→ $41.8B

t-center AI spend, 2025 → 2032 (Omdia) — the buyer this lab designs for.

gap

and cancellation are the same market read twice. The six patterns in this lab are the controls that decide which side of it a deployment la

ecks its own claims before anyone acts on them.

urvived verification · survival rate 0.769

: 0.60 = GUESS (flagged, not hidden)

exactly the 3 flagged (0.42 / 0.35 / 0.48) — in the demo this holds by construction; the les

utcome and the headline moves to 0.692 — try it live in the notebook (§ "Change one outc

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POC 01 · CTO CONCLUSIONS

An Agent That Sees

Grounded perception + a verification loop

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

13 claims

region-grounded, each with a confidence

10 / 13 survived

survival rate 0.769 against ground truth

3 flagged = 3 failed

every GUESS (below 0.60) failed the check — by construction in the demo

0 hidden

low confidence is flagged, never buried

THE CTO CONCLUSION

Perception without a verification loop is unpriced risk.
The survival rate is the price tag — demand it before scale.

Run it as a measured pilot: your scenes, human checkers, one site.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

e system of record — coordination you can replay.

4 tasks · 2 agents · 0 side channels

78.00 unused gym charge (Feb–Mar) · coffee subscription billed 3x in March vs 2x · secon

replay and audit read it alone

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POC 02 · CTO CONCLUSIONS

Conversation Is a Coordination Primitive

The transcript is the system of record

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

12 messages
the entire collaboration — nothing outside the transcript

4 analysis tasks
coordinated by two agents with no shared memory

3 unasked-for findings
incl. a $78 unused charge the analyst never queried

100% replayable
audit = read the transcript; there is nowhere else for state to hide

THE CTO CONCLUSION

If agent work cannot be replayed from one record,
you do not have automation — you have anecdotes.

Run it as a measured pilot: one workflow, transcript-of-record on.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

contract, linted before the agent ever runs.

s (Role · Output · Do-Nots · Signal)

nt CLEAN · Bad contract: 3 flags (MISSING Do-Nots, WEAK_SIGNAL, VAGUE_OUTPUT)

gations · termination on the typed signal EXTRACTION COMPLETE (check returned SIGNA

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POC 03 · CTO CONCLUSIONS

RODS — Contract Before Capability

Four lines, linted before the agent runs

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

4 lines

Role · Output · Do-Nots · Signal — the whole contract

3 flags

on the bad contract: missing Do-Nots, vague output, weak signal

0 flags

on the good contract — it runs; the other never executes

1 typed signal

EXTRACTION COMPLETE ends the run — never a prose word

THE CTO CONCLUSION

Contract failures are preventable before runtime.
Four lines and a checker price the risk before deployment.

Adopt it as the intake gate: no contract, no run.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synthetic data only

and the label teaches the next attempt.

SUCCESS — outcomes computed from the run record (runs/attempts.csv), never typed in

the recovery habit — print the schema first — is attempt 2, and the revised query succeed

rong (category 'gym' matches no rows); the exact name from the totals ('gym-unused') su

(2 x $39.00, 2026-02 and 2026-03), traced to rows in data/transactions.csv (22 rows)

finding moves — the perturbation test moves the gym amounts and watches $78.00 be

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POC 04 · CTO CONCLUSIONS

The Error Is the Curriculum

Five attempts, typed outcomes, one finding traced to the rows

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

5 attempts

every one typed — 1 ERROR · 1 EMPTY · 3 SUCCESS, from the run record

2 failure kinds

ERROR = the code was wrong · EMPTY = the question was — routed apart

$78.00 found

an unused subscription, 2 x $39.00, traced to the actual rows

1 recovery habit

print the schema first — the error became the next attempt's input

THE CTO CONCLUSION

An agent's failures are its training data.
Price the loop by how it recovers, not by its first try.

Adopt it where answers must explain themselves: type every failure.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synth

count what each shape costs in calls.

omparison.csv: round robin 6 agent turns / 6 total calls · selector 3 agent turns + 4 routin

thopaedics, 2 slots — Tue 09:30, Wed 14:10 — unbooked, awaiting human confirmation)

d robin runs its sequence twice by construction; the swarm takes a verification hand-back

OMPLETE (tests/test_perturbation.py, failed-router case)



e human review to the closest fights.

l at +8% — REJECT (strength 5 for / 11 against) · C2 24-hour incident-notice duty — PROC

gument in the script and the reviewer's first stop

ompt for human review: it locates no genuine uncertainty, is not a risk register, and a rea

counts are computed, not typed (tests/test_perturbation.py)

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POC 06 · CTO CONCLUSIONS

Disagreement Is a Sensor

Structured debate that rations human review — and decides nothing

KEY DATA POINTS · SCRIPTED DEMONSTRATION (ILLUSTRATIVE)

3 positions

contract clauses debated from the same evidence, both ways

2 REJECT · 1 PROCEED

verdicts computed from the strength of each side's case

0.62

the closest fight — the 24-hour notice duty; the reviewer's first stop

0 decisions

made by the script: it allocates review; humans and counsel decide

THE CTO CONCLUSION

Review time is the scarcest control in governance.
Spend it where the argument is hardest to call.

Adopt it as the review queue: both cases on record, closest first.

Full notebook, code and evidence: github.com/GAI-Engineering/poc-learning-lab · MIT / CC BY 4.0 · Synth

· POC 03 (RODS)

ern is the most transferable asset in the lab: four lines, a linter, a typed completion signal. It generalises to every agent the lab will ever sh

02 (transcripts)

ord pattern proceeds behind its durability gates: persistence, replay and failure recovery proven before any stronger claim.

no — POC 01 (perception)

g artefact until a real perception adapter replaces the scripted scene. Its lesson — the verification loop — is already load-bearing in POC 0

04–06)

utor loop, group-chat routing shapes and structured debate extend the same discipline: labelled outcomes, counted calls, disagreement t

METHOD — HOW THIS LAB EARNS TRUST

ta files in the repo — change an input and the result moves; each notebook carries a perturbation test that proves it.

the work: INTERIM notebooks are scripted demonstrations; a FINAL promotion requires a real recorded run, reproducible end to end

ndent evaluator of record, with conditions recorded in RELEASES.md and closed only on evidence.

awn SVG infographics in business/infographics/; generated images are checked figure-by-figure at full size before they are used.

n the commit history — including this deck’s predecessors.